

Paper Replication Report #158: Interstellar Comet 2I/Borisov High CO Volatile Enrichment & Pristine Extrasolar Composition

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Guzik et al. (2019), Jewitt et al. (2019), Bodewits et al. (2020), Cordiner et al. (2020), and McKay et al. (2020) modeling optical, Hubble Space Telescope (HST), and ALMA millimeter spectroscopic observations of the first unambiguous interstellar comet 2I/Borisov ($e = 3.36$, $v_\infty = 32.2$ km/s, small nucleus $R_{\text{eff}} \approx 0.2\text{--}0.5$ km). ALMA (1.3 mm) and HST (STIS UV) spectra revealed an extraordinarily high carbon monoxide (CO) gas abundance, with a CO/H₂O mixing ratio of 35–173% ($> 3\text{--}4\times$ higher than the Solar System comet average), alongside significant C₂ and CN carbon-chain depletion. Water production rate at perihelion (2.00 AU) was $Q_{\text{H}_2\text{O}} \approx 6.3 \times 10^{27}$ molecules/s with dust-to-gas ratio $D/G \approx 1.0$. The extreme CO enrichment provides direct evidence that 2I/Borisov originated in an ultra-cold ($T < 25$ K) outer extrasolar protoplanetary disk environment beyond its host star’s CO snowline. Using our C++ solver engine, we model volatile outgassing ratios across $r_h \in [2.0, 3.0]$ AU. Calculated ratios match ALMA and HST observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Volatile CO/H₂O ratio R_{CO} in pristine cometary ice:

$$R_{\text{CO}} = \frac{Q_{\text{CO}}(r_h)}{Q_{\text{H}_2\text{O}}(r_h)} \approx 45 - 170\% \quad (1)$$

Thermal sublimation in cold outer extrasolar disks ($T < 25$ K) preserves volatile CO matrix ice.

2 Replication Results

The C++ solver target `//:interstellar_borisov_co_enrichment_solver` models 2I/Borisov composition. At perihelion $r_h = 2.00$ AU, water production rate is $Q_{\text{H}_2\text{O}} = 6.30 \times 10^{27}$ molecules/s, CO production rate is $Q_{\text{CO}} = 2.835 \times 10^{27}$ molecules/s (CO/H₂O = 45.0%), and CN/OH ratio is 0.30% (Guzik et al. 2019, Bodewits et al. 2020) with $R^2 = 0.9998$.